

Qualification Pack



Printed Circuit Board Technician

QP Code: NIE/ELE/Q0903

Version: 1.0

NSQF Level: 4

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP
Pocket, Sector-8, Dwarka
New Delhi-110077 || email:puneetnema@nielit.gov.in

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NIE/ELE/Q0903: Printed Circuit Board Technician

Brief Job Description

This course provides an overview of Printed Circuit Board (PCB) technology, starting with an exploration of basic concepts such as electronic component packages, types of PCBs, and the history of PCB development. Participants gain insight into Electronic Design Automation (EDA) tools, with a focus on customizing Autodesk Eagle for in-depth analysis and design. The curriculum delves into component types, including through-hole packages, and emphasizes adherence to design rules. Participants learn the CAD workflow and the importance of manufacturing documentation. Additionally, they develop proficiency in creating PCB flowcharts, understanding layering and materials, and simulating electronic circuits to analyze waveform outputs. Furthermore, the course covers Film Master generation methods, material selection, and assembly techniques for both leaded and surface-mounted devices, along with familiarization with assembly tools.

Personal Attributes

A Printed Circuit Board (PCB) Technician needs strong attention to detail and technical aptitude to handle intricate designs and electronic components. They should possess good problem-solving skills for troubleshooting circuit issues and have manual dexterity for precise soldering and component handling. Patience and focus are essential, as the work often involves complex and time-consuming tasks. Strong analytical thinking is required to test and diagnose circuit functionality effectively. Additionally, good communication and adaptability are important for working with teams and keeping up with technological advances.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0914: Fundamentals of PCB Design](#)
2. [NIE/ELE/N0915: Printed Circuit Board Design and Analysis Techniques](#)
3. [NIE/ELE/N0916: PCB Manufacturing and Assembly Techniques](#)
4. [NIE/ELE/N0913: Implementation of PCB Design Project](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
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Sub-Sector	Electronics Assembly and Services
Occupation	Electronics Assembly, Electronics Equipment Services
Country	India
NSQF Level	4
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015 / 8212.2701 , Electrical and Electronics Equipment Assemblers
Minimum Educational Qualification & Experience	12th Class OR Completed 2nd year of 3-year diploma (after 10th) and pursuing regular diploma OR 10th grade pass plus 2-year NTC OR 10th grade pass with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (3) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/07/2027
NSQC Approval Date	25/07/2024
Version	1.0
Reference code on NQR	QG-04-EH-02854-2024-V2-NIELIT
NQR Version	2

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NIE/ELE/N0914: Fundamentals of PCB Design

Description

This module focuses on the design and analysis of Printed Circuit Boards (PCBs), emphasizing their role as vital components in electronic systems. It differentiates PCBs from PWBs, highlighting PCB's complexity and integration with integrated circuits. Learners will explore various types of PCBs, such as Single-Sided and Multi-Layer, and materials like FR-4, understanding their impact on performance and cost. The module covers Electronic Design Automation (EDA), including the history and current trends, and introduces EDA tools for design, analysis, and optimization. Key tools include SPICE, PSpice for simulation, and Autodesk Eagle for efficient PCB design. Additionally, learners will understand different PCB components and package types, including through-hole packages, and their role in PCB development.

Scope

The scope covers the following :

- The NOS for Printed Circuit Board Design and Analysis Techniques encompasses understanding PCBs' critical role in electronic systems, distinguishing them from PWBs, and exploring various types and materials like FR-4. It includes learning about Electronic Design Automation (EDA), its history, current trends, and tools such as SPICE, PSpice, and Autodesk Eagle for PCB design and simulation. The scope also covers identifying different PCB components and package types, including through-hole packages. Key outcomes include proficiency in PCB design using Autodesk Eagle, differentiation between PCB and PWB, and effective use of EDA tools for circuit design and analysis.

Elements and Performance Criteria

Introduction to Printed Circuit Board

To be competent, the user/individual on the job must be able to:

- PC1.** Understand PCB as a vital electronic component for interconnecting electronic parts.
- PC2.** Differentiate between PCB and PWB, noting PCB's complexity and integration with integrated circuits.
- PC3.** Recognize types of PCBs (Single Sided, Multi-Layer) and materials used (e.g., FR-4), and understand their impact on performance and cost.

Electronic design Automation (EDA)

To be competent, the user/individual on the job must be able to:

- PC4.** Learn about the history of EDA and current market trends, emphasizing its importance in modern electronics development.
- PC5.** Explore various EDA tools and their functionalities for design, analysis, and optimization of electronic circuits.
- PC6.** Get introduced to SPICE and PSpice for circuit simulation, and understand their role in design validation and optimization.
- PC7.** Familiarize with AUTODESK Eagle for PCB design, learning its interface, tools, and workflow for efficient project implementation.

Component and their categories and Package Types

To be competent, the user/individual on the job must be able to:

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- PC8.** Understand the different kinds of components used in PCB design and as well as the categories of the components that are used to do the development of the PCB board.
- PC9.** Acquire the knowledge of the through-hole packages.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the essential role of Printed Circuit Boards (PCBs) in electronic systems, their distinction from Printed Wiring Boards (PWBs), and the impact of different PCB types (e.g., Single-Sided, Multi-Layer) and materials (e.g., FR-4) on performance, reliability, and cost.
- KU2.** Gain insight into the history and current trends of EDA tools, their functionalities for circuit design, analysis, and optimization, and become proficient with tools such as SPICE, PSpice, and Autodesk Eagle for PCB design and simulation.
- KU3.** Recognize various PCB components and their categories, and understand different package types, including through-hole packages, their characteristics, and their role in PCB development.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop expertise in using Electronic Design Automation (EDA) tools, including Autodesk Eagle, SPICE, and PSpice, for designing, simulating, and optimizing PCBs. Acquire the ability to create accurate and efficient PCB layouts and perform detailed circuit analysis.
- GS2.** Cultivate strong analytical skills to evaluate and differentiate between PCB types and materials, and understand their implications for performance and cost. Apply problem-solving techniques to optimize circuit designs and troubleshoot issues during the design and manufacturing processes.
- GS3.** Enhance the ability to effectively communicate technical concepts related to PCB design and analysis. Prepare clear and detailed documentation for PCB designs, including component specifications, design rationale, and simulation results, to ensure clarity and facilitate collaboration with team members and stakeholders.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Printed Circuit Board</i>	10	7	-	-
PC1. Understand PCB as a vital electronic component for interconnecting electronic parts.	-	-	-	-
PC2. Differentiate between PCB and PWB, noting PCB's complexity and integration with integrated circuits.	-	-	-	-
PC3. Recognize types of PCBs (Single Sided, Multi-Layer) and materials used (e.g., FR-4), and understand their impact on performance and cost.	-	-	-	-
<i>Electronic design Automation (EDA)</i>	20	8	-	-
PC4. Learn about the history of EDA and current market trends, emphasizing its importance in modern electronics development.	-	-	-	-
PC5. Explore various EDA tools and their functionalities for design, analysis, and optimization of electronic circuits.	-	-	-	-
PC6. Get introduced to SPICE and PSpice for circuit simulation, and understand their role in design validation and optimization.	-	-	-	-
PC7. Familiarize with AUTODESK Eagle for PCB design, learning its interface, tools, and workflow for efficient project implementation.	-	-	-	-
<i>Component and their categories and Package Types</i>	20	8	-	-
PC8. Understand the different kinds of components used in PCB design and as well as the categories of the components that are used to do the development of the PCB board.	-	-	-	-
PC9. Acquire the knowledge of the through-hole packages.	-	-	-	-
NOS Total	50	23	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0914
NOS Name	Fundamentals of PCB Design
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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NIE/ELE/N0915: Printed Circuit Board Design and Analysis Techniques

Description

This module covers key aspects of PCB design and analysis, starting with footprint design and analysis. Learners will understand the essential rules for PCB design, the workflow of computer-aided design (CAD) tools, and the importance of manufacturing documents. It includes an overview of PCB design flow charts and the various layers and materials used in PCB construction. Students will also gain practical skills in Autodesk Eagle, learning to simulate both basic and analog electronic circuits, analyze power supplies for efficient power delivery, and master simulation techniques for sensor modules and digital circuits.

Scope

The scope covers the following :

- The NOS for Printed Circuit Board Design and Analysis Techniques covers essential aspects of PCB design, including footprint design and the importance of adhering to design rules and CAD workflows. It involves understanding the PCB design flow chart, various layers, and materials used in PCB construction. Learners will gain practical skills in Autodesk Eagle, including simulating basic and analog circuits, analyzing power supplies for efficient delivery, and mastering simulation techniques for sensor modules and digital circuits. The scope emphasizes creating accurate manufacturing documents and implementing effective design strategies to ensure successful PCB development.

Elements and Performance Criteria

Study of Foot Print Design & Analysis

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the rules before PCB Designing.
- PC2.** Understanding the flow of computer aided design packages.
- PC3.** Acquire the importance of manufacturing documents.

PCB Designing Flow Chart and Layers

To be competent, the user/individual on the job must be able to:

- PC4.** Understand the designing of flow chart of PCB board.
- PC5.** Understand the various layers and materials used to design the PCB. Acquire the basic knowledge

Simulation on AutoDesk Eagle

To be competent, the user/individual on the job must be able to:

- PC6.** Understand and simulate basic and analog electronic circuits.
- PC7.** Analyze power supplies to ensure efficient power delivery.
- PC8.** Master simulation techniques for various sensor modules and digital circuits.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Understand the key rules and guidelines for PCB footprint design, the workflow of computer-aided design (CAD) tools, and the importance of accurate manufacturing documents for successful PCB production.
- KU2.** Learn about the PCB design process, including the design flow chart, and gain knowledge of the various layers and materials used in PCB construction, and their impact on design and performance.
- KU3.** Develop skills in simulating basic and analog circuits, analyzing power supplies, and validating sensor and digital circuit designs using Autodesk Eagle to ensure efficient and effective PCB design.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop advanced skills in using CAD tools and Autodesk Eagle for PCB design and simulation, including accurately creating footprints, comprehending design workflows, and conducting detailed circuit analysis.
- GS2.** Develop strong analytical skills to assess and interpret PCB design rules, materials, and simulation results, ensuring effective problem-solving and optimization in the design process.
- GS3.** Enhance the ability to produce and understand detailed manufacturing documents, communicate technical information clearly, and collaborate effectively with team members throughout the PCB design and development process.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Study of Foot Print Design & Analysis</i>	15	7	-	10
PC1. Understand the rules before PCB Designing.	-	-	-	-
PC2. Understanding the flow of computer aided design packages.	-	-	-	-
PC3. Acquire the importance of manufacturing documents.	-	-	-	-
<i>PCB Designing Flow Chart and Layers</i>	20	8	-	5
PC4. Understand the designing of flow chart of PCB board.	-	-	-	-
PC5. Understand the various layers and materials used to design the PCB. Acquire the basic knowledge	-	-	-	-
<i>Simulation on AutoDesk Eagle</i>	15	7	-	5
PC6. Understand and simulate basic and analog electronic circuits.	-	-	-	-
PC7. Analyze power supplies to ensure efficient power delivery.	-	-	-	-
PC8. Master simulation techniques for various sensor modules and digital circuits.	-	-	-	-
NOS Total	50	22	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0915
NOS Name	Printed Circuit Board Design and Analysis Techniques
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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NIE/ELE/N0916: PCB Manufacturing and Assembly Techniques

Description

This module focuses on the key techniques used in PCB manufacturing and assembly. Learners will explore both traditional film-based and modern digital techniques for translating electronic designs into physical layouts, with a strong emphasis on accurate layout creation. Critical manufacturing steps such as plating, etching, and cleaning will be covered to ensure reliable board production. The module emphasizes real-world applications, including quality control in manufacturing. In PCB assembly, learners will gain foundational knowledge of assembly techniques for both leaded and surface-mounted devices, as well as the tools used in the process. Additionally, the module includes a detailed study of soldering techniques, exploring the materials and methods involved in creating secure and efficient connections on PCBs.

Scope

The scope covers the following :

- The scope of this module covers the key techniques in PCB manufacturing and assembly. It includes learning traditional and modern methods for converting electronic designs into physical layouts, focusing on critical manufacturing steps like plating, etching, and cleaning to ensure accuracy and reliability. The module also introduces assembly techniques for leaded and surface-mounted devices, emphasizing the tools used during the process. Additionally, students will gain knowledge of soldering methods and materials to ensure proper connections, with a focus on applying these skills in real-world manufacturing and quality control scenarios.

Elements and Performance Criteria

PCB Manufacturing Techniques

To be competent, the user/individual on the job must be able to:

- PC1.** Learn both traditional film and modern digital techniques for PCB design, focusing on converting electronic designs into physical layouts and understanding precise layout creation methods.
- PC2.** Gain insight into critical steps of PCB manufacturing, including plating, etching, and cleaning techniques to ensure accurate and reliable board production.
- PC3.** Apply acquired knowledge to real-world manufacturing scenarios, emphasizing quality control for consistent and reliable results.

PCB Assembly Techniques

To be competent, the user/individual on the job must be able to:

- PC4.** Understand the basic level knowledge required to understand assembly techniques for leaded and Surface mounted devices.
- PC5.** Acquire the basic level knowledge of use of various tools during assembly.

Study Soldering Techniques

To be competent, the user/individual on the job must be able to:

- PC6.** Understand the methods of soldering of PCBs, material used in soldering process.
- PC7.** Understand methods of soldering

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand traditional and modern methods for turning electronic designs into physical layouts, including key processes like plating, etching, and cleaning, and learn how to ensure accuracy and reliability in PCB production with effective quality control.
- KU2.** Gain foundational knowledge of assembly methods for leaded and surface-mounted devices (SMD), along with an understanding of the tools needed for precise and efficient component assembly.
- KU3.** Learn the techniques and materials used for soldering PCBs, focusing on proper methods to ensure strong, reliable electrical connections in both manual and automated processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop proficiency in PCB manufacturing processes, including plating, etching, cleaning, and quality control, to ensure accurate and consistent production of PCBs.
- GS2.** Gain practical skills in assembling leaded and surface-mounted devices (SMDs) using the appropriate tools, ensuring precision and efficiency in the assembly process.
- GS3.** Master soldering techniques for both manual and automated processes, ensuring strong, reliable electrical connections on PCBs while maintaining high standards of workmanship.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>PCB Manufacturing Techniques</i>	35	15	-	10
PC1. Learn both traditional film and modern digital techniques for PCB design, focusing on converting electronic designs into physical layouts and understanding precise layout creation methods.	-	-	-	-
PC2. Gain insight into critical steps of PCB manufacturing, including plating, etching, and cleaning techniques to ensure accurate and reliable board production.	-	-	-	-
PC3. Apply acquired knowledge to real-world manufacturing scenarios, emphasizing quality control for consistent and reliable results.	-	-	-	-
<i>PCB Assembly Techniques</i>	35	15	-	5
PC4. Understand the basic level knowledge required to understand assembly techniques for leaded and Surface mounted devices.	-	-	-	-
PC5. Acquire the basic level knowledge of use of various tools during assembly.	-	-	-	-
<i>Study Soldering Techniques</i>	30	15	-	5
PC6. Understand the methods of soldering of PCBs, material used in soldering process.	-	-	-	-
PC7. Understand methods of soldering	-	-	-	-
NOS Total	100	45	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0916
NOS Name	PCB Manufacturing and Assembly Techniques
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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NIE/ELE/N0913: Implementation of PCB Design Project

Description

This National Occupational Standard (NOS) outlines the essential skills and knowledge required for successful design and fabrication. It emphasizes proficiency in using PCB design software, adhering to industry standards, and ensuring quality control throughout the process. Key competencies include the ability to analyze project requirements, create schematic diagrams, and develop PCB layouts. Effective collaboration with cross-functional teams and troubleshooting design issues are also critical components. Lastly, the NOS highlights the importance of staying updated with technological advancements and best practices in PCB design.

Scope

The scope covers the following :

- The scope of the National Occupational Standard (NOS) for PCB Design implementation encompasses the complete design process, from initial concept to final layout, ensuring adherence to specifications. It includes proficiency in using industry-standard PCB design software for schematic capture and layout creation. Quality assurance methodologies are emphasized to validate PCB designs for reliability and performance. Effective collaboration with cross-functional teams is essential for ensuring manufacturability and project success.

Elements and Performance Criteria

Design and Implementation of PCB

To be competent, the user/individual on the job must be able to:

- PC1.** Acquire the basic level knowledge required to understand Film Master Generation method, material used for manufacturing, cleaning methods of base material.
- PC2.** Understand the methods for manufacturing of PCBs. Type of material used for manufacturing. Understand the mechanical method in manufacturing.
- PC3.** Understand the methods of soldering of PCBs. material used in soldering process. Understand Methods of soldering.
- PC4.** Understand the basic concept of SMD Soldering and understand repair and rework methods.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the basic concept of thermal analysis and understanding how to design PCB for manufacturing and assembly point of view.
- KU2.** Acquire the basic level knowledge required to understand assembly techniques for leaded and SMDs. Acquire the basic level knowledge of use of various tools during assembly.
- KU3.** Understand the design and manufacturing Techniques of Printed Circuit Boards.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Develop expertise in using Electronic Design Automation (EDA) tools, including Autodesk Eagle, SPICE, and PSpice, for designing, simulating, and optimizing PCBs. Acquire the ability to create accurate and efficient PCB layouts and perform detailed circuit analysis.
- GS2.** Cultivate strong analytical skills to evaluate and differentiate between PCB types and materials, and understand their implications for performance and cost. Apply problem-solving techniques to optimize circuit designs and troubleshoot issues during the design and manufacturing processes.
- GS3.** Enhance the ability to effectively communicate technical concepts related to PCB design and analysis. Prepare clear and detailed documentation for PCB designs, including component specifications, design rationale, and simulation results, to ensure clarity and facilitate collaboration with team members and stakeholders.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design and Implementation of PCB</i>	-	-	50	10
PC1. Acquire the basic level knowledge required to understand Film Master Generation method, material used for manufacturing, cleaning methods of base material.	-	-	-	-
PC2. Understand the methods for manufacturing of PCBS. Type of material used for manufacturing. Understand the mechanical method in manufacturing.	-	-	-	-
PC3. Understand the methods of soldering of PCBs. material used in soldering process. Understand Methods of soldering.	-	-	-	-
PC4. Understand the basic concept of SMD Soldering and understand repair and rework methods.	-	-	-	-
NOS Total	-	-	50	10

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0913
NOS Name	Implementation of PCB Design Project
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

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6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0914.Fundamentals of PCB Design	50	23	0	0	73	18
NIE/ELE/N0915.Printed Circuit Board Design and Analysis Techniques	50	22	0	20	92	25
NIE/ELE/N0916.PCB Manufacturing and Assembly Techniques	100	45	0	20	165	35
NIE/ELE/N0913.Implementation of PCB Design Project	0	0	50	10	60	12
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	220	120	50	50	440	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

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Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf